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Data Interactions

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SET09123 – Interactive Data Visualisation

Lecture Summary

- Interactivity with Visualisations
- Interacting with Data
- Dimensionality Reduction

Interactivity

Motivation

Why would you want an interactive visualisation?

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Why would you want an interactive visualisation?

- To overcome human and machine limitations
- To deal with *complex and large* datasets
  - A visualisation that changes can display different facets of the data
- To support *several user tasks*
  - A visualisation that responds to user inputs will be more engaging and useful

Types of Changes

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There are two ways to change or update a visualisation

- Change the Data
- Change the View

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- **Change the Data**
- Change the View

Data Interactions

Interacting with Data?

- We want to overcome human and machine limitations
- We want to facilitate user tasks over large datasets, e.g.:
  - Searching within multiple ranges of attribute values
  - Querying a hierarchical subset
  - Exploring a specific spatial region in more details
- We must handle data on the fly to respond to user inputs

Typical Mode: Reduction

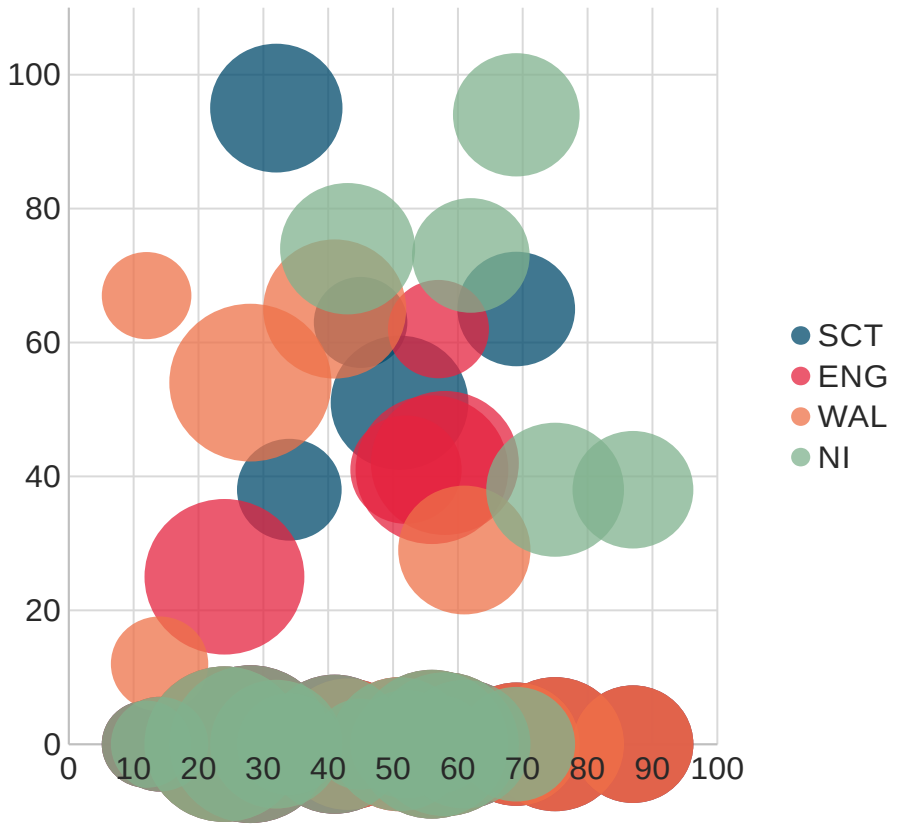
- Typically, the data is mostly **reduced** to make the visualisation *simpler* or *more focused* on an area of interest

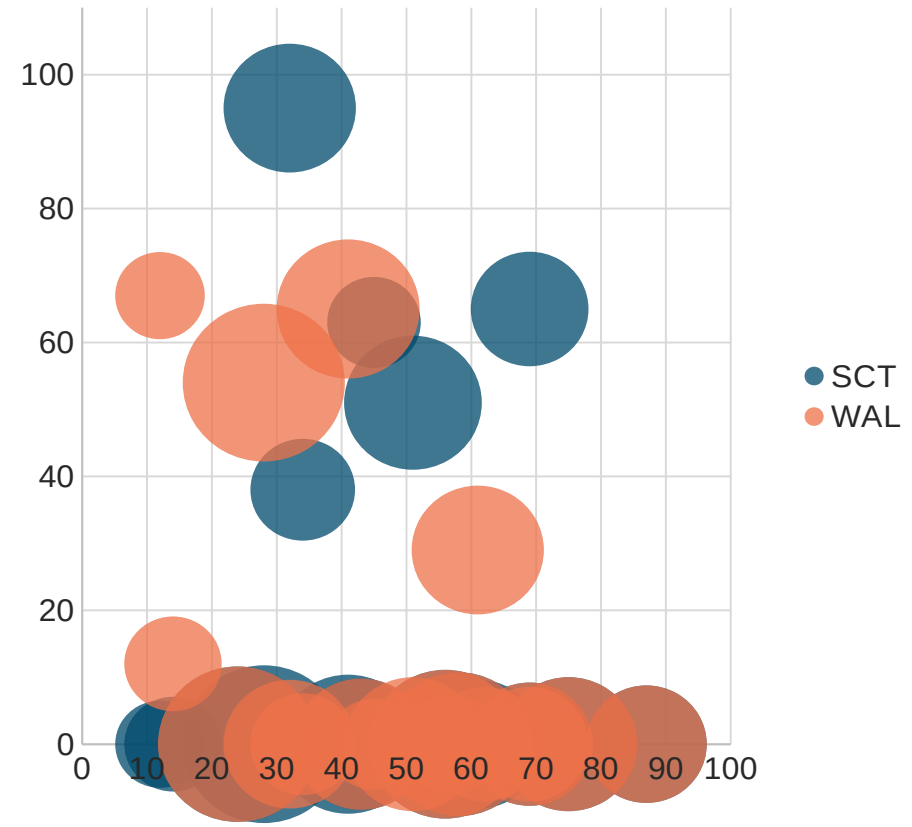
- In some instances, the data is *reduced by default* and user inputs enable the data augmentation of specific regions

Reduction by Filtering

- To **remove** items or attributes from the presented data
- Filtering items: only present a subset of items that fall within a selected range of attributes' values
- Filtering attributes: only show visual encoding for a selected set of attributes

Filtering Items





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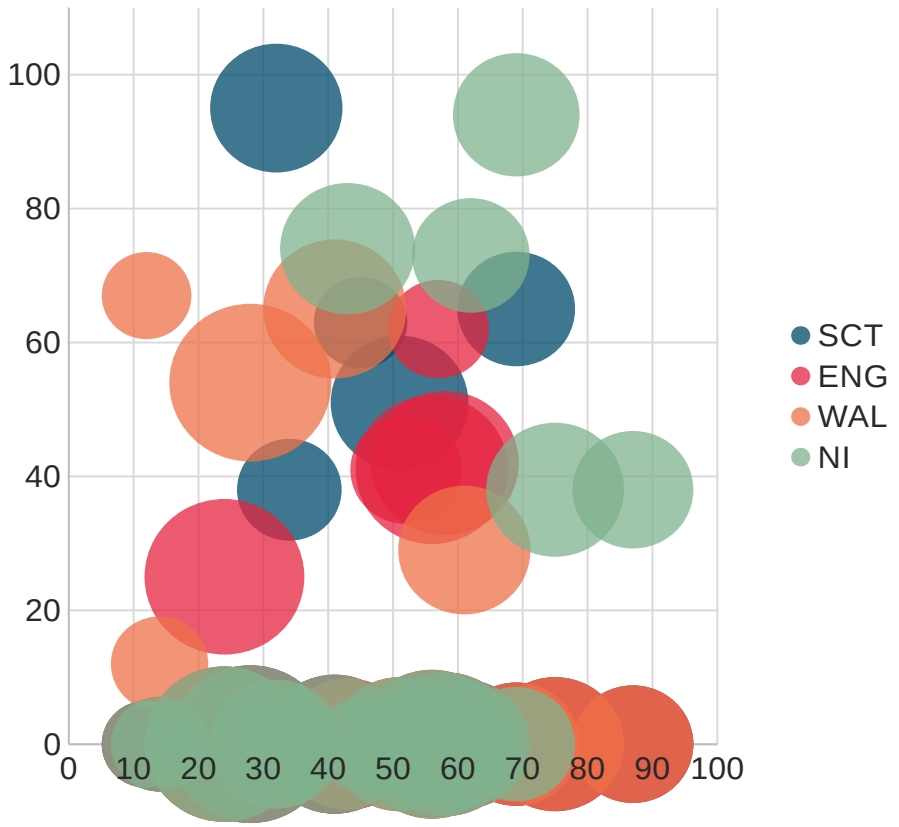
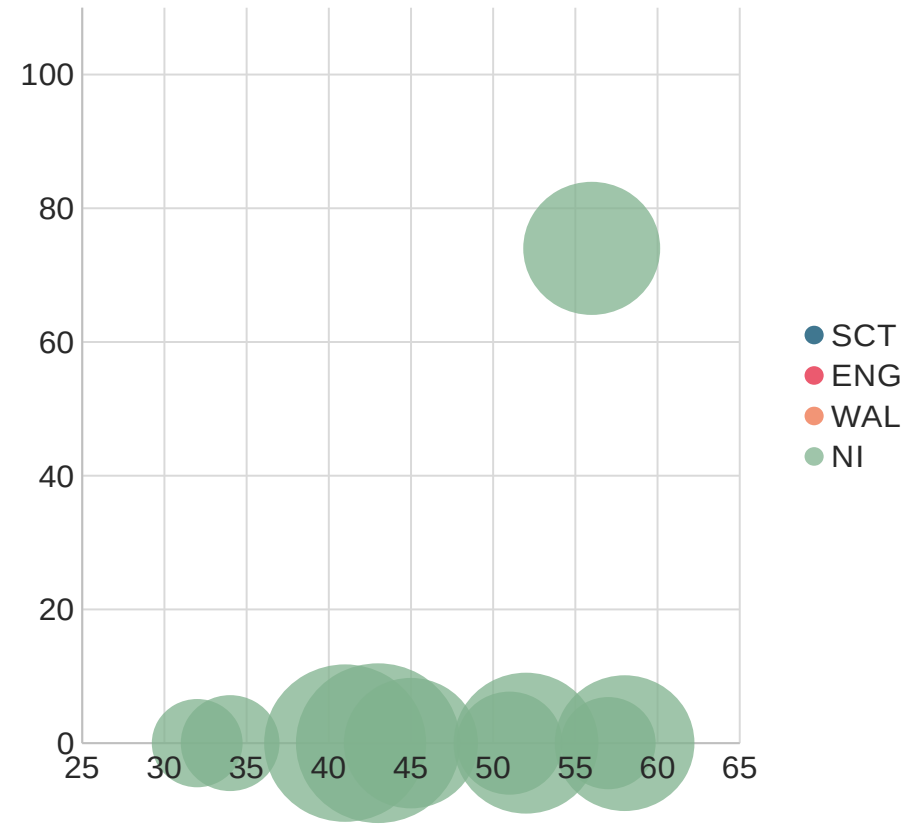
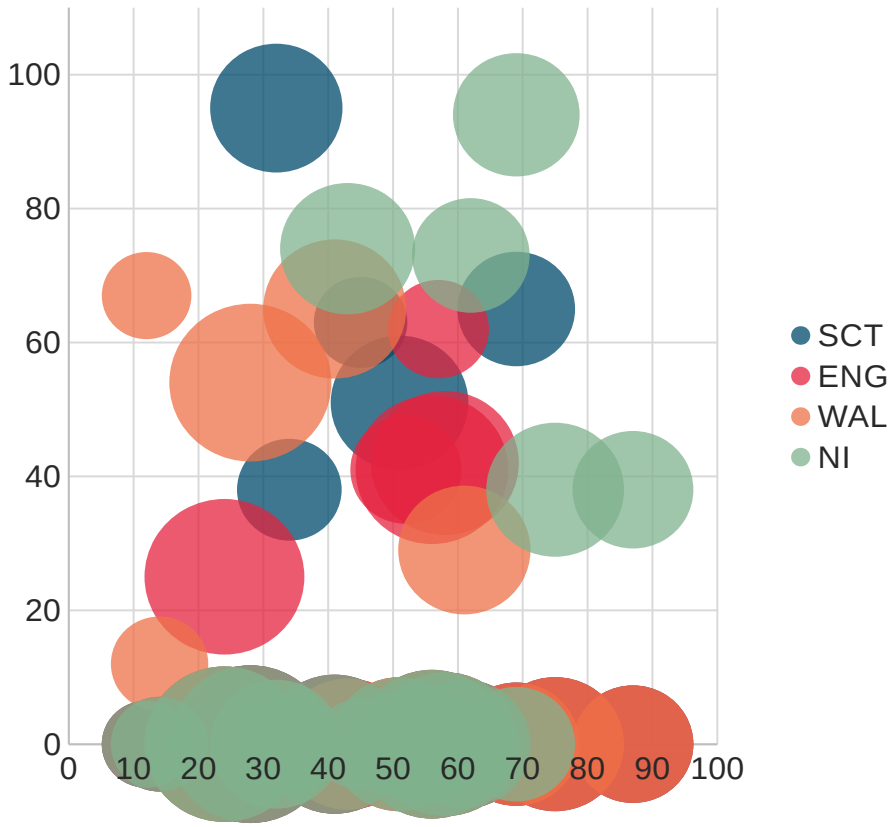
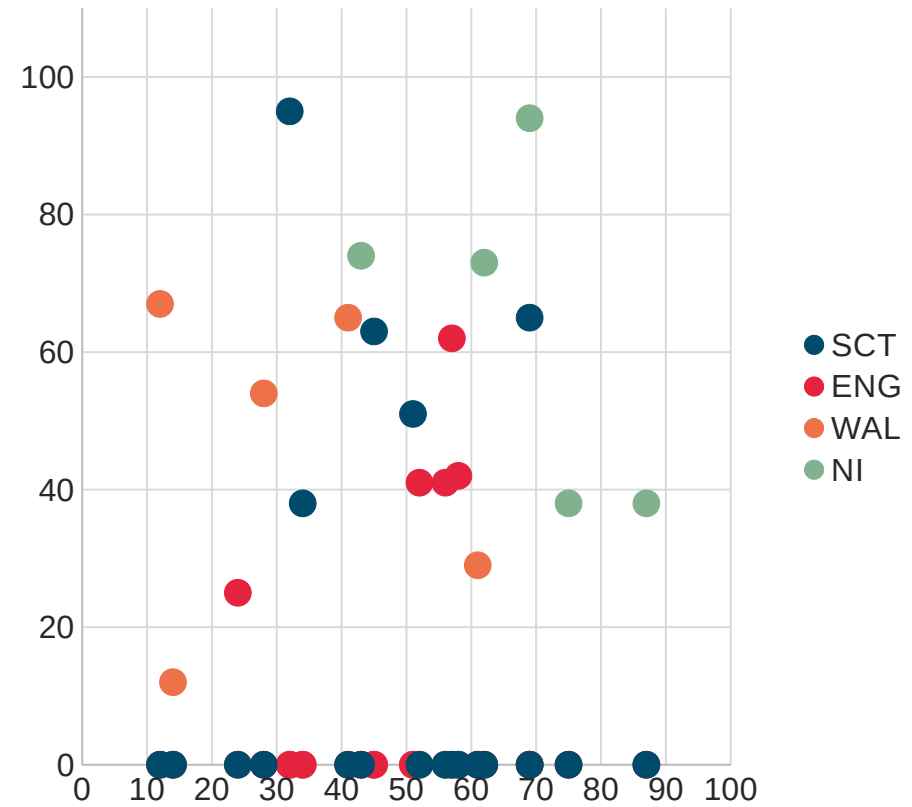




Filtering Attributes





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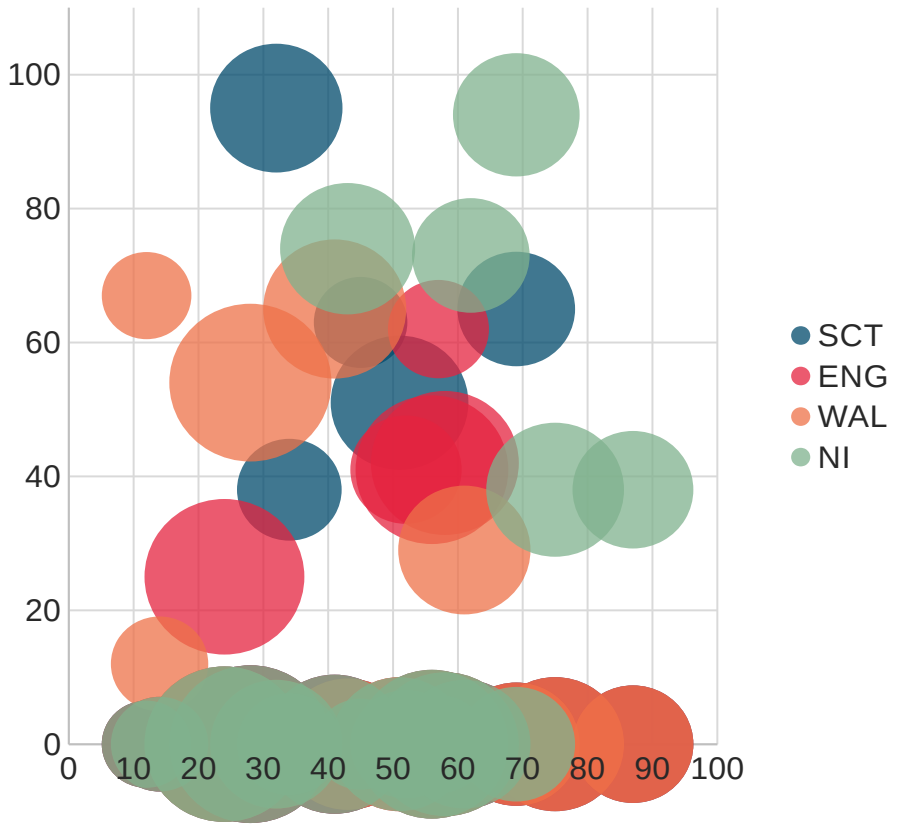

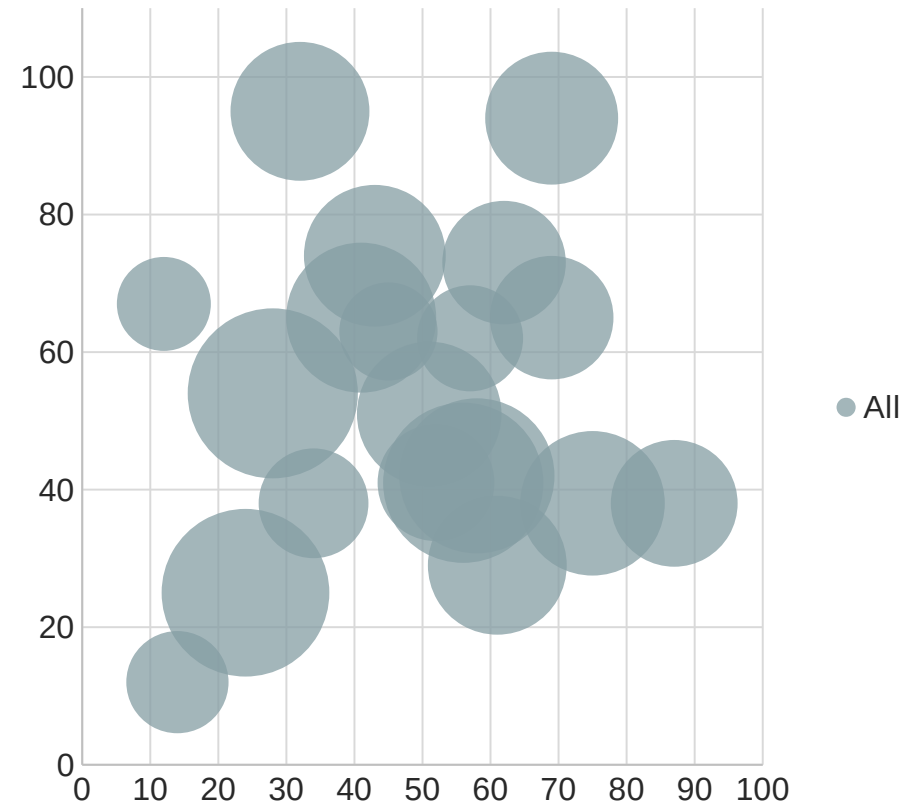

Reduction by Aggregation

- To **merge** groups of items or attributes into a new single derived element that will represent the group
- Aggregating items: creating an abstract item that covers a certain range of attributes' values
- Aggregating attributes: synthesising new attributes to replace the original attributes

Aggregating Items

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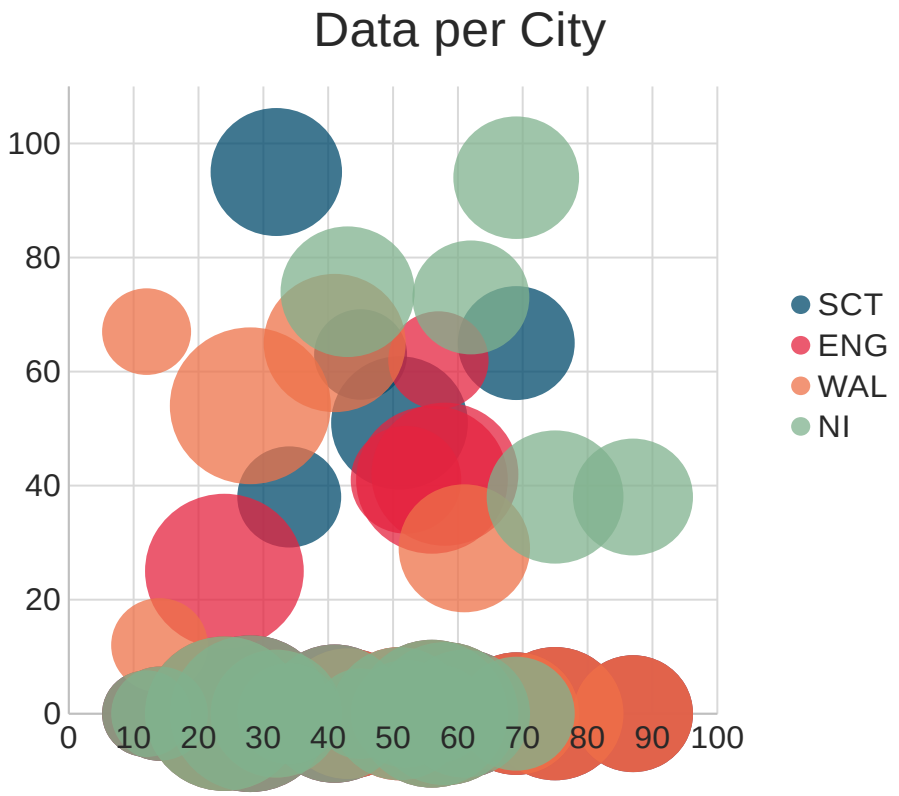
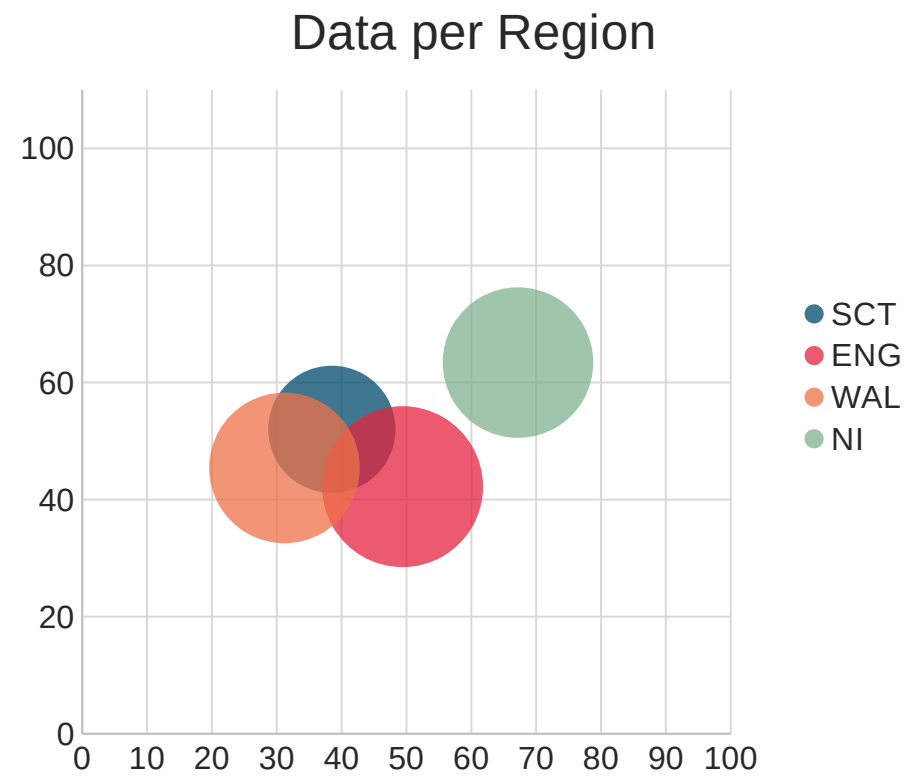
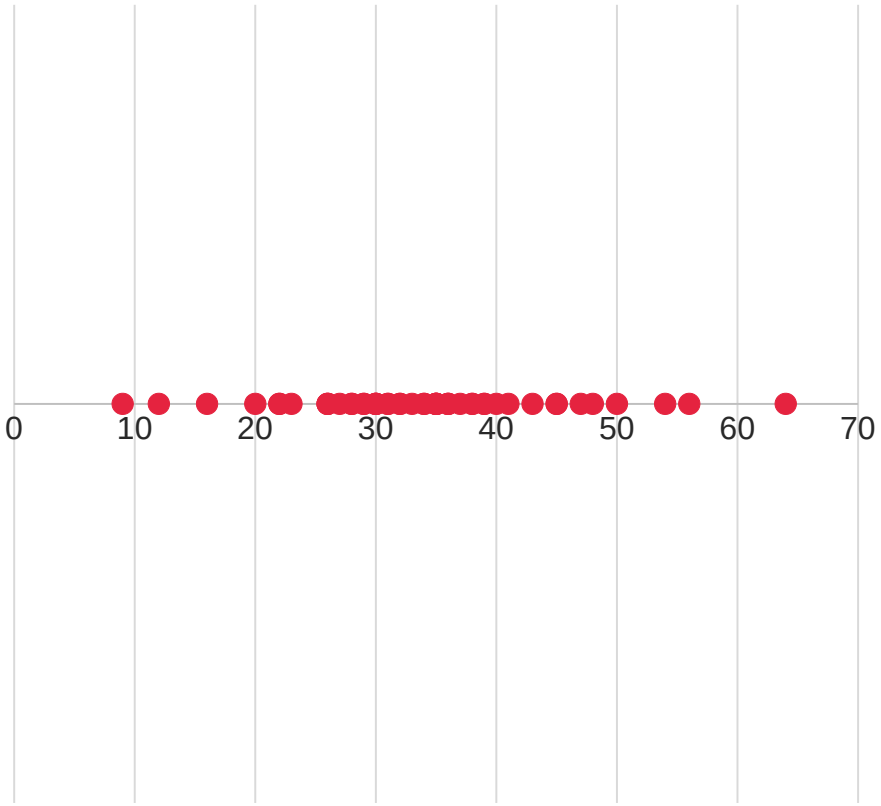
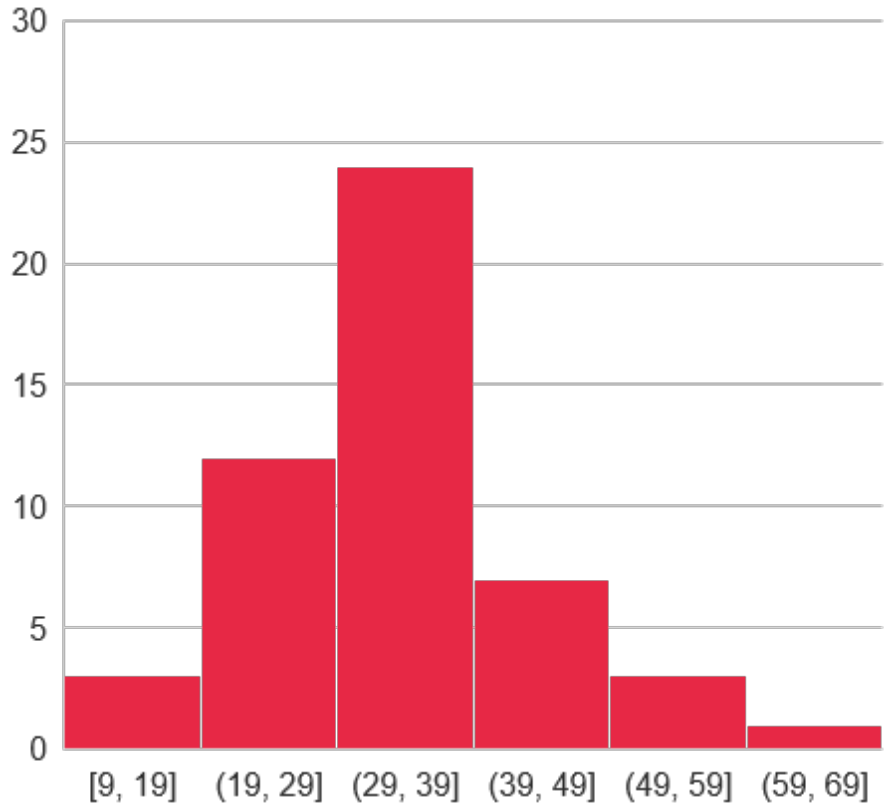


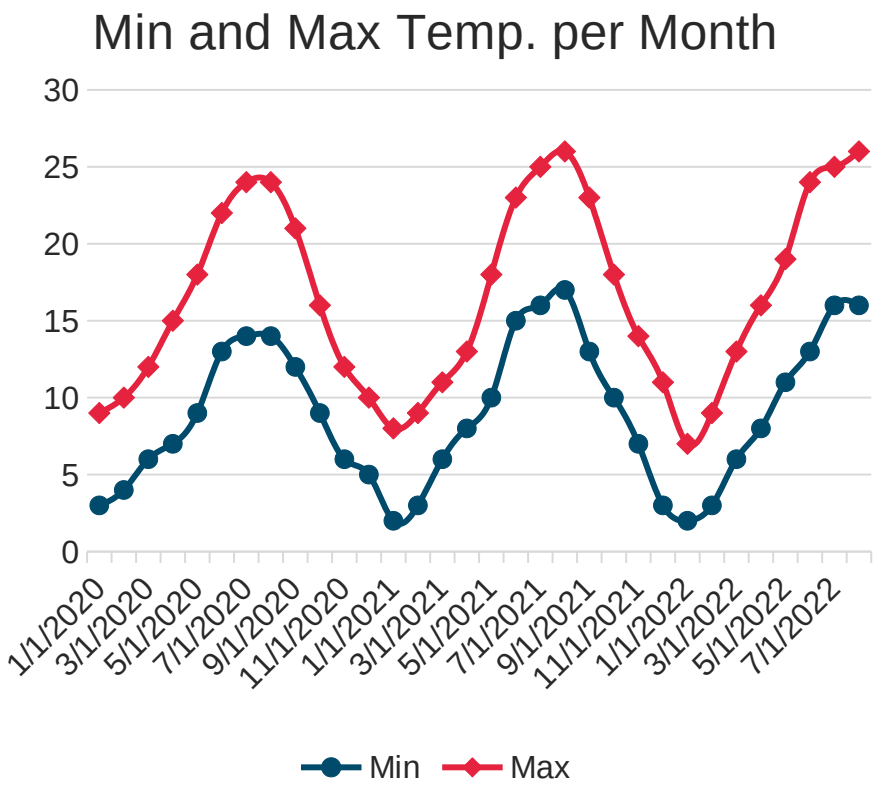


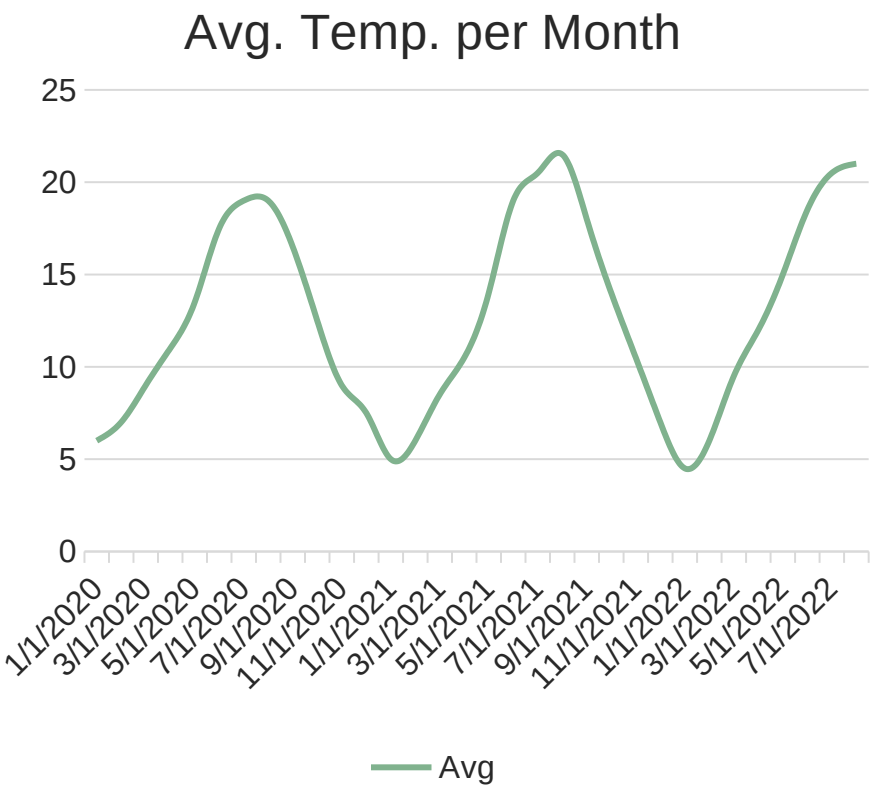
Aggregating Items





Aggregating Attributes





Aggregating Attributes

- Grouping and averaging attributes works when they have the same unit
- Otherwise we need to use algorithms that *reduce dimensionality*

Interaction is Bidirectional!

- It is important to let users get a simpler and more focused view
- It is also important to let them see or go back to the big picture
- Hence, un-filtering and de-aggregating data interactions need to be programmed too

Dimensionality Reduction

A Brief Intro

Dimensionality Reduction

- Reducing the number of dimensions in a dataset

Attr 1	Attr 2	Attr 3	…	Attr N
23	98	67	…	96
26	84	73	…	31
78	54	32	…	74
…	…	…	…	…

X	Y
15	36
14	32
21	45
…	…

Dimensionality Reduction

- Reducing the number of dimensions in a dataset
- While preserving the dataset "*structure*"

Attr 1	Attr 2	Attr 3	…	Attr N
23	98	67	…	96
26	84	73	…	31
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Dimensionality Reduction

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There are two approaches:

- Feature selection – selecting the attributes best representing all others (filtering)
- Feature extraction – calculating new representative attributes (aggregating)

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- **Feature extraction** – calculating new representative attributes (aggregating)

Feature Extraction

- Linear – the lower dimensional dataset is a linear transformation of the original dataset
  - The key is finding a plane of best fit, where the data can be best preserved
- Nonlinear – the original dataset is a nonlinear manifold
  - When the data is “curved” and a plane wouldn’t work

Linear Techniques

- Principle Component Analysis (PCA)
  - Finds the combinations of dimensions with most variance, i.e., the principle components of the data
- Latent Discriminant Analysis (LDA)
  - Similar to PCA but aims to improve the separation between classes (supervised)

Nonlinear Techniques

- Multi-Dimensional Scaling (MDS)
  - Aims to best preserve the pair-wise Euclidean distances between points
- Isometric Mapping (ISOMAP)
  - Aims to best preserve the pair-wise geodesic distances between points
- t-Stochastic Neighbour Embedding (t-SNE)
  - Aims to best preserve local structures

- To learn more:

Anowar, F., Sadaoui, S., & Selim, B. (2021).

Conceptual and empirical comparison of dimensionality reduction algorithms (PCA, KPCA, LDA, MDS, SVD, LLE, ISOMAP, LE, ICA, t-SNE).

Computer Science Review, 40, 100378.

https://doi.org/10.1016/j.cosrev.2021.100378

Conclusion

Key Points

- Part of the interaction paradigms rely on interacting with the dataset to create refined visualisations

- The data can be reduced (or augmented) by means of:
  - **Filtering** – removing items or attributes from the view
  - **Aggregation** – merging items or attributes into abstract representatives

- Dimensionality Reduction is a class of algorithms designed to represent data in lower-dimensional spaces